The Omics Hub

Command Line & Bioinformatics Cheat Sheet

1. Command Line Basics

Navigation & Files

- `pwd` : Print working directory (where am I?)
- `ls -lh` : List files with human-readable sizes
- `cd /path/to/dir` : Change directory
- `cp f1 f2` : Copy file1 to file2
- `mv old new` : Move or rename a file
- `rm file` : Remove a file (CAUTION)

Examining Data

- `head -n 10 data.fq` : View first 10 lines
- `tail -n 10 data.fq` : View last 10 lines
- `less data.fq` : View file page-by-page (`q` to exit)
- `wc -l file.txt` : Count lines in a file

Searching & Filtering

- `grep "motif" file.fa` : Find lines with "motif"
- `grep -c "motif" file.fa` : Count occurrences
- `grep -v "motif" file` : Lines WITHOUT motif
- `awk '{print $1, $3}'` : Print 1st and 3rd columns

2. HPC & Environment

Job Management (Slurm)

- `squeue -u user` : Check status of your jobs
- `scancel 123456` : Cancel job ID 123456
- `sbatch script.sh` : Submit a batch job

Example Slurm Header

```
#!/bin/bash
#SBATCH --job-name=scRNA_analysis
#SBATCH --nodes=1
#SBATCH --cpus-per-task=8
#SBATCH --mem=32G
#SBATCH --time=24:00:00
#SBATCH --output=analysis_%j.log
```

Conda / Mamba

- `conda create -n env python=3.9` : New env
- `conda activate env` : Activate environment
- `conda install -c bioconda fastqc` : Install

3. Single-Cell & Immunology Marker Genes

Key biomarkers for T-cell malignancies, Sézary Syndrome, and general immune profiling.

Gene / Marker	Cell Type & Functional Relevance
CD4 / CD8	Defines major T-cell lineages: Helper T cells (CD4+) and Cytotoxic T cells (CD8+).
FOXP3	Master transcriptional regulator for Regulatory T cells (Tregs). Critical for distinguishing suppressive phenotypes.
CD25 (IL2RA)	Alpha chain of the IL-2 receptor. Highly expressed on Tregs (often FOXP3+CD25+) and activated T cells.
CLIC1	Chloride Intracellular Channel 1. Key marker associated with metabolic vulnerabilities in malignant cells.
CCR4 / CCR7	Chemokine receptors essential for T-cell skin homing (CCR4) and lymph node migration (CCR7).
PTPRC (CD45)	Pan-leukocyte marker, expressed on all nucleated hematopoietic cells.